MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 5939A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specificaiton MIL-E-1.

DESCRIPTION: Window (pre-TR when used in pairs, see note 2), frequency range 1,250 to 1,350 MHz, nominal peak rating 575 kw

ABSOLUTE RATINGS:

Parameter:	Transmitter po	Altitude
Unit:	kw	ft
Maximum:	575	10,000
Minimum:	10	---

PHYSICAL CHARACTERISTICS:

Dimensions: See figure 1
Mountings: See Drawing 210-JAN
Packaging: See MIL-E-75 and figure 6

TEST CONDITIONS:

Parameter:	Transmitter po	tpc	Prr	VSWR
Unit:	kw	µs	pps	---
Maximum:	575	4.6	330	1.10:1
Minimum:	525	3.4	270	---

Frequency		
F	MHz	±
F1	1,250	1
F2	1,286	1
F3	1,300	1
F4	1,350	1

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
4471	Recovery time	8	F = F1 and F4	t	---	36	µs
1031	High-frequency vibration	---		---	---	---	---
	Quality conformance inspection, part 1	1					
4416	Insertion loss (1) (fixed tuned)	6	Fo = F2	Li	---	0.30	dB
4471	Recovery time	8	F = F2	t	---	36	µs
1027	Temperature cycling (nonoperating)	---		---	---	---	---
---	Temperature cycling end point:						
---	High-level operation	4,7	F = F2	ΔT	---	110	°C
---	Resonant frequency	3,5	VSWR = 1.05 (max)	F	1,280	1,292	MHz
	Quality conformance inspection, part 2						
4416	Insertion loss (2) (fixed tuned)	6	VSWR = 1.05 (max): F = F1 F = F3 and F4	 Li Li	 --- ---	 0.60 0.50	 dB dB
1126	Glass envelope strain	---		---	---	---	---
(G) 4452	Leakage power	7	VSWR = 1.05 (max); F = F2; tpc = 0.4 µs	Po	---	0.026	Joules
4473	Low-level VSWR	5	VSWR = 1.05 (max): F = F1 and F4 F = F3	 --- ---	 --- ---	 1.4 1.2	 --- ---
4488	Arc loss	---	Fo = F2	ΔLi	---	0.3	dB
---	High-level operation	4,7	F = F2	ΔT	---	110	°C
	Qualification conformance inspection, part 3						
(G) ---	Life test	4,7	Group C; F = F2	t	500	---	hrs
---	Life test end points:						
(G) 4471	Recovery time	8	Group C; F = f2	t	---	45	µs
(G) 4552	Leakage power	7	Group C; F = F2; tpc = 0.4 µs	Po	---	0.032	Joules
(G) 1027	Temperature cycling life test	---	Group C	---	10	---	Cycles
---	Temperature cycling life-test end point:						
(G) ---	High-level operation	4,7	Group C; F = F2	ΔT	---	110	°C

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. For pairs of windows, separation 2.938 ±0.063 inches (74.63 ±1.60 mm) between centerlines.
3. Resonant frequency corresponds to a minimum standing wave produced by the window. The resonant frequency may be measured by any equivalent method.
4. Test circuit "C" shall be used with branch line open (without attenuator and bolometer mount). Temperature rise shall not exceed the specified value after reaching equilibrium. There shall be no evidence of sparking.
5. Use test circuit A, figure 5.
6. Use test circuit B, figure 3.
7. Use test circuit C, figure 4.
8. Use test circuit D, figure 2.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 99
DLA - ES

User activities:
Army - WC
Navy - AS, OS, CG
Air Force - 11, 17

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3166)

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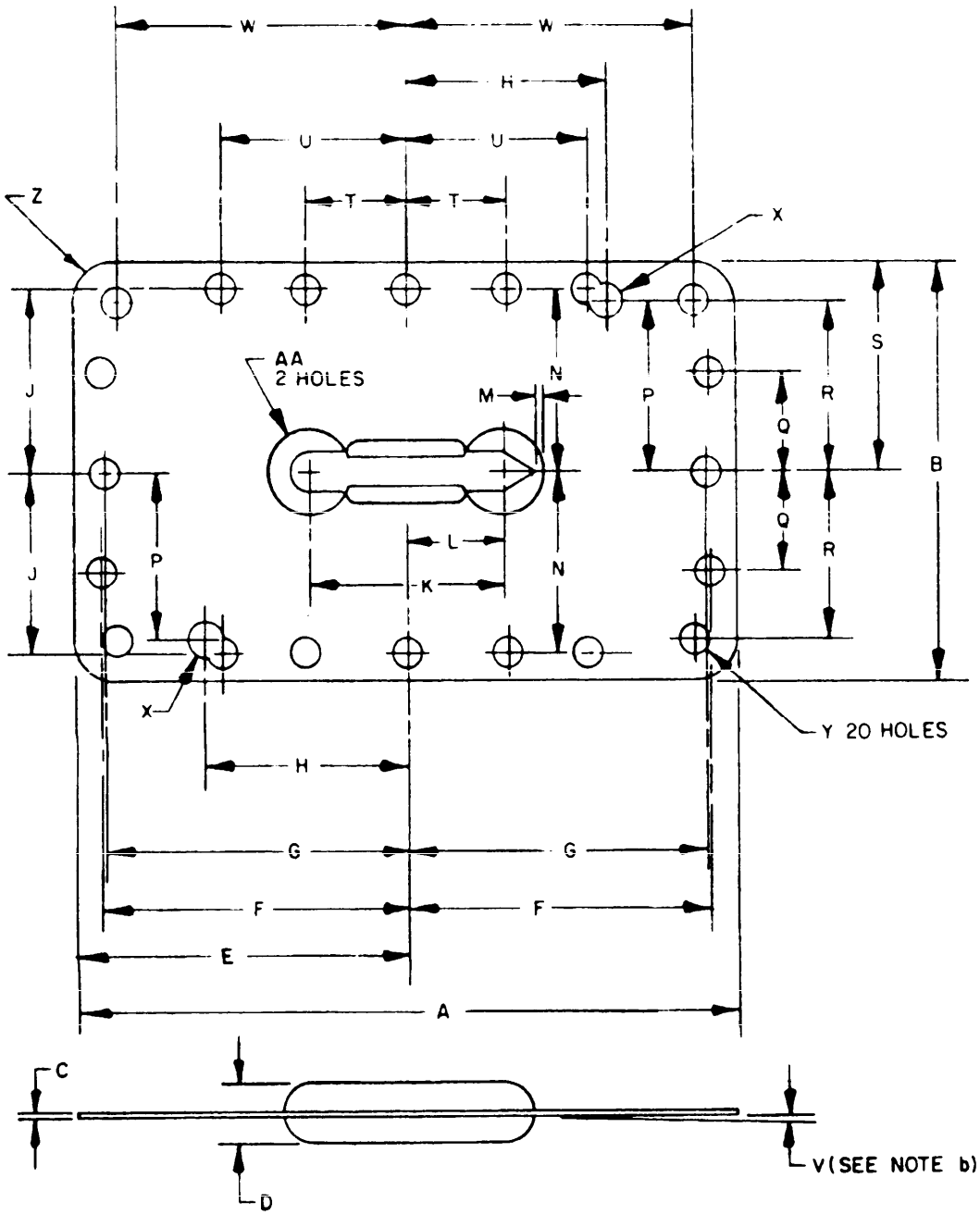


FIGURE 1. Outline drawing of electron tube type 5939A.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
A	8.485 (215.52)	8.515 (216.28)
B	5.235 (132.97)	5.265 (133.73)
D	.375 (9.53)	
E	4.235 (107.57)	4.265 (108.33)
K	2.500 (63.50)	
L	1.250 (31.75)	
S	2.610 (66.29)	2.640 (67.06)
Z	.485R (12.32)	.515R (13.08)
AA	1.000 (25.40)	
Quality conformance inspection, part 1		
C	.059 (1.50)	.067 (1.70)
M	.063 (1.60)	
V		.047 (1.19)
Quality conformance inspection, part 2		
F	3.922 (99.62)	3.952 (100.38)
G	3.891 (98.83)	3.921 (99.59)
H	2.593 (65.86)	2.623 (66.62)
J	2.298 (58.37)	2.328 (59.13)
N	2.266 (57.56)	2.296 (58.32)
P	2.079 (52.81)	2.109 (53.57)
Q	1.235 (31.37)	1.265 (32.13)
R	2.110 (53.59)	2.140 (54.36)
T	1.286 (32.66)	1.316 (33.43)
U	2.360 (59.95)	2.390 (60.71)
W	3.734 (94.84)	3.763 (95.58)
X	.389 Dia (9.88)	.393 Dia (9.98)
Y	.367 Dia (9.32)	.383 Dia (9.73)

NOTES:
a. Six holes located by dimensions J and U and horizontal centerline.
b. Permissible bow.
c. Four holes located by dimensions N and T.
d. Four holes located by dimensions R and W.
e. Mounting plate shall be plated in accordance with QQ-N-290A (class 1).
f. Four holes located by dimensions F and Q.
g. Two holes located by dimensions G and vertical centerline.
h. Two holes located by dimensions H and P.

FIGURE 1. Outline drawing of electron tube type 5939A - Continued

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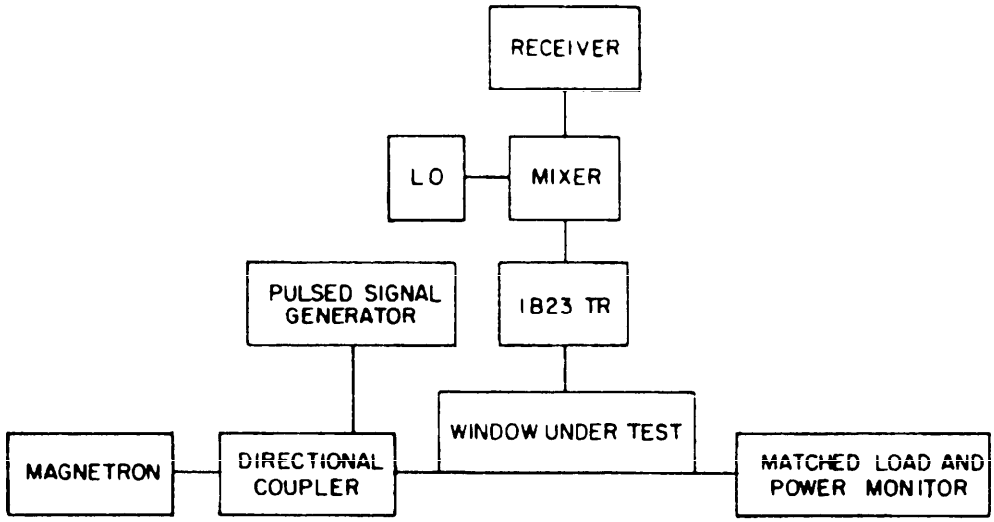


FIGURE 2. Test circuit D.

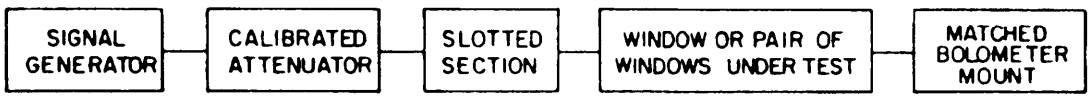


FIGURE 3. Test circuit B.

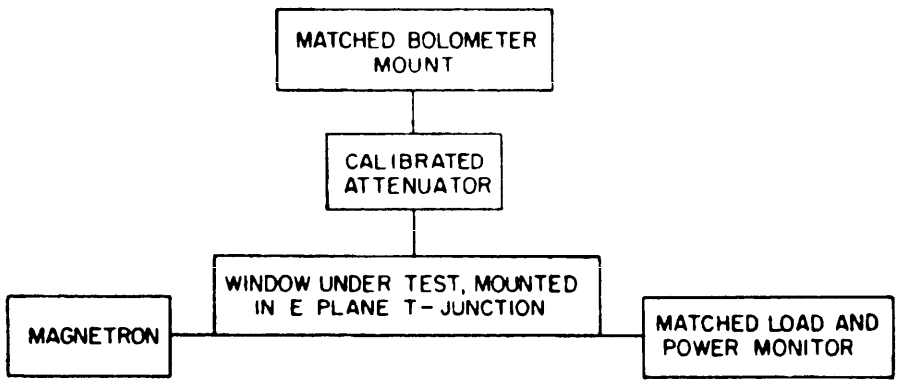


FIGURE 4. Test circuit C.

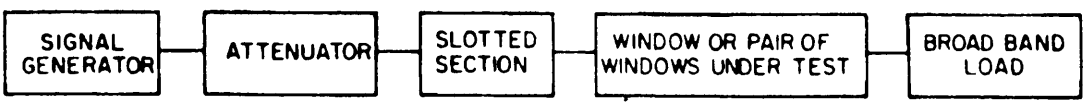


FIGURE 5. Test circuit A.

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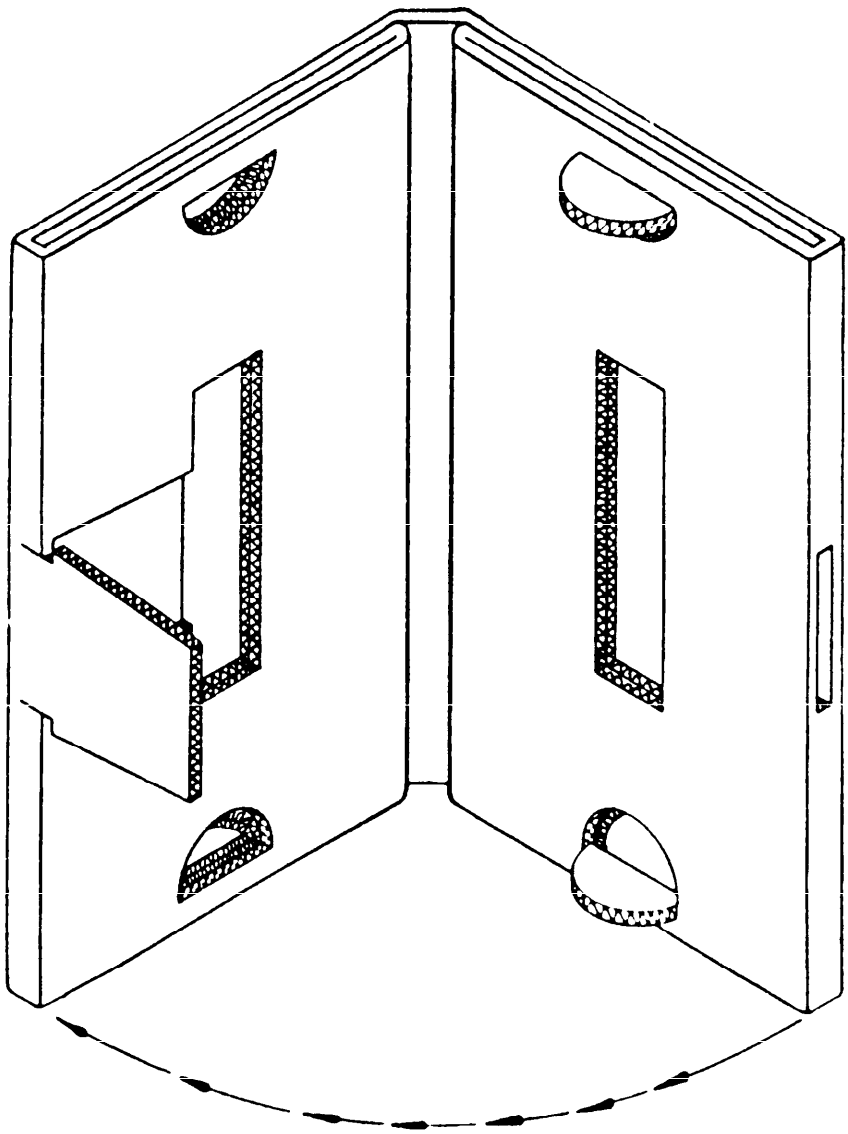
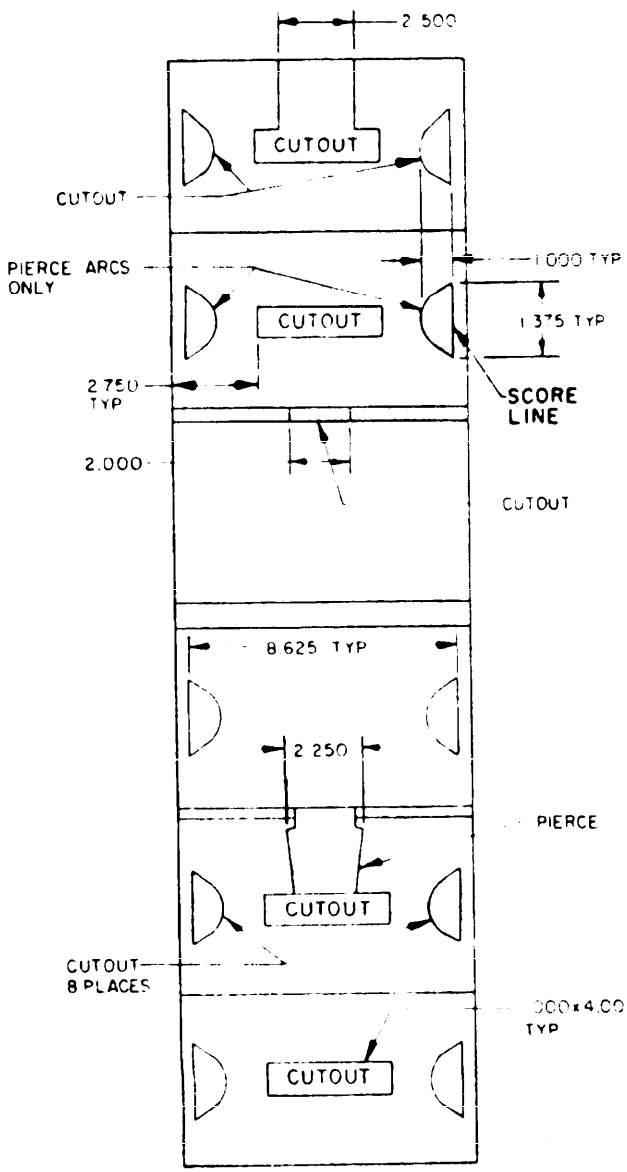


FIGURE 6. Packaging requirements.

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INCHES	MM
1.000	25.40
1.375	34.93
2.000	50.80
2.250	57.15
2.500	63.50
2.750	69.85
4.000	101.60
8.625	219.08

NOTES:
a. Dimensions are in inches.
b. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
c. Cutouts 1.00 x 4.00 shall be in alinement and be central to insert.
d. Ears on tab shall be located so that insert is locked when closing tab is inserted in cutout.
e. Each unit shall be enclosed in a folded corrugated wrapper as illustrated, and the folded wrapper shall then be enclosed in a water-vaporproof bag.
f. Twenty units are packed in a domestic shipping container 9-3/4 (247.65) x 9-1/2 (241.30) x 11-1/2 (292.1[illegible]).

FIGURE 6. Packaging requirements - Continued

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